

Experiment: Ensemble Methods (Bagging, Boosting, and Stacking)

1. Setup Environment

- Open Python Notebook (Jupyter/Colab).
- Import required libraries: scikit-learn, pandas, matplotlib, numpy.

2. Load Dataset

- Select a dataset (e.g., Iris, Breast Cancer, or any suitable classification dataset).
- Identify features and target variable.

3. Preprocess Data

- Handle missing values if any.
- Normalize or standardize features if required.
- Split dataset into training and testing sets.

4. Bagging (Bootstrap Aggregating)(Random Forest)

- Initialize a base learner (e.g., Decision Tree).
- Train multiple models on bootstrap samples of the dataset.
- Aggregate predictions (majority vote for classification, average for regression).
- Evaluate accuracy and analyze variance reduction.

5. Boosting

- Initialize a base learner (weak classifier, e.g., shallow Decision Tree).
- Train the model sequentially, giving more weight to misclassified samples.
- Combine weak learners into a strong classifier.
- Evaluate performance improvement compared to single model.
- Explore algorithms such as AdaBoost or Gradient Boosting.

6. Stacking (Stacked Generalization)

- Train multiple diverse base models (e.g., Decision Tree, SVM, Logistic Regression).
- Generate predictions from base models.
- Use a meta-model (e.g., Logistic Regression) to combine base model outputs.
- Train the meta-model on predictions of base models.
- Evaluate overall performance against individual models.

7. Model Evaluation

- Compare accuracy, precision, recall, F1-score of Bagging, Boosting, and Stacking.
- Use confusion matrices and ROC-AUC for further comparison.

8. Visualization

- Plot comparison charts of performance metrics across Bagging, Boosting, and Stacking.
- Visualize feature importances where applicable.

9. Experiment with Parameters

- For Bagging: change number of estimators (trees).
- For Boosting: vary learning rate and number of weak learners.
- For Stacking: try different combinations of base learners and meta-models.

10. Summarize Results

- Report best-performing ensemble method.
- Highlight strengths of Bagging (variance reduction), Boosting (bias reduction), and Stacking (model combination).
- Discuss trade-offs between accuracy, training time, and interpretability.